

PAPER_1348 — Hubbard Model Mott Transition

Framework: UQFF — Star-Magic v5.27+ | **Tier:** F Condensed Matter (CLOSED) **Calculator:**
calculate_paradox({"paradox": "hubbard_model"})

Master Expression

$U/t = D_{\text{phys}} = 4$ EXACT integer-primitive

Match: within obs 4-8

Interpretation

Mott insulator transition driven by D_{phys} threshold. F_{TRZ} -modulated hopping suppression.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$
- $\omega_{\text{SCm}} = 1.25$ THz

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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